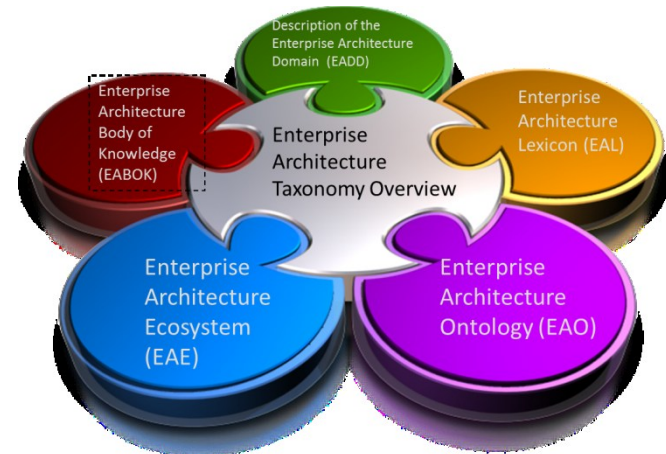


FEAPO Member Organizations Call for Glossary entries

Prepared by FEAPO Taxonomy WG

November, 2012



Pursuing the FEAPO EA Lexicon

- The Enterprise Architecture Lexicon (EAL) elaborates the essential taxonomy within the scope of EA to identify the richness of terms and relationships between terms commonly used by practitioners to articulate EA artifacts. This work product is closest to the commonly held notion of an EA domain of discourse.
- Chose to use Formal Concept Analysis and graphical techniques to examine glossary terms from Taxonomy WG participant organizations.
- Objective is identification of lexicon, i.e. the vocabulary we use to characterize EA, and the primary relationships we assign to those words in term definitions.

EA Lexicon Work Plan

1. Capture terms from EADD
2. Circulate terms among members seeking additional EA terms and linkage between terms
3. Create hyper-graph of terms for review and discussion
4. Identify foundation terms of general usage for agreement
5. Prune hyper graph to foundation terms and strong associations
6. Prepare EA Lexicon and partial grammar
7. Submit EA Lexicon to Council for acceptance by FMO's

Taxonomy WG EADD Participants

- International Council on Systems Engineering (INCOSE)
- Institute of Electrical and Electronics Engineers (IEEE)
- Data Management Association (DAMA)
- Business Architecture Guild / Business Architecture Society (Guild/BAS)
- International Organization for Standardization (ISO)

Capture terms from EADD

- Now have 10+ glossary definitions from 5 organizations
 - INCOSE, IEEE, DAMA, Guild/BAS, ISO
 - 62 terms
 - 451 distinct vocabulary candidates
- Prune terms and vocabulary to focus on shared use
- Identify shared concepts using techniques of Formal Concept Analysis (based on work of Ganter & Wille)
 - Formal Concept Analysis helps us identify the relationships among our use of terms across the domains of the professional organizations participating as FEAPO members.
 - FCA result is lattice of concepts
- We need more FMO participants to extend lexical reach!

Formal Concept Analysis

- FCA recognizes that a **formal context** consists of a set of **objects** (terms in our situation) and a set of **attributes** (definitions in our situation) and an **incidence relation** that maps the objects with attributes.
- Task is applying FCA by crafting the attributes such that the formal mechanisms of FCA are useful.
- Requires consistent expression of objects and attributes across the domain glossaries.
- We need the essence of the descriptive definition associated with the term, not the well-crafted statement.
- The procedure outlined below should produce a suitable set of objects and attributes for each domain.

Preparing for FCA

- Begin with a Glossary, Definitions list, or similar itemization of terms adopted by your professional organization. Select terms that best articulate your concepts for the domain of enterprise architecture.
- Re-write the definition of each term using descriptive words
 - present tense verbs, singular nouns, and no prepositions or pronouns
 - short noun or verb phrases, 2 or 3 words, with hyphens between words are acceptable
 - keep term and definition adjectives and adverbs distinct
 - do not add words that are not already in the definition or derived directly from the words in the definition

Map relations in FCA context

- In an Excel spreadsheet, enter the distinct terms, your FCA objects, as separate columns and the distinct descriptive words, your FCA attributes, as separate rows.

	term1	term2	term-3
descriptive1	X		
descriptive2		X	X
descriptive3		X	
descriptive4	X		
descriptive-5			X

- The formal context of your professional organization's enterprise architecture domain is now ready for analysis and comparison with the other EA domain contexts.

ISO Term examples

- Organization – structure of the distribution of responsibilities and authorities in the enterprise
term := organization
descriptors := structure, distribution, responsibility, authority, enterprise
- Architecture – conceptualization of the form, function, and fitness-for-purpose of a system in its environment, as embodied in the elements of the system, the relationships between those elements, the relationship of the system to its environment and the principles guiding the design and evolution of the system
term := architecture
descriptors := conceptual, form, function, fit-for-purpose, system, environment, embody, element, relationship, principle, design, evolution

Business Architecture examples

- Organization – a social unit of people, systematically structured and managed to meet a need or to pursue collective goals on a continuing basis.

term := organization

descriptors := social-unit, people, systematic, structure, manage, meet, need, pursue, collective, goals, continue

(note that 'basis' is considered as part of 'continue')

- Business Architecture – a blueprint of the enterprise that provides a common understanding of the organization and is used to align strategic objectives and tactical demands

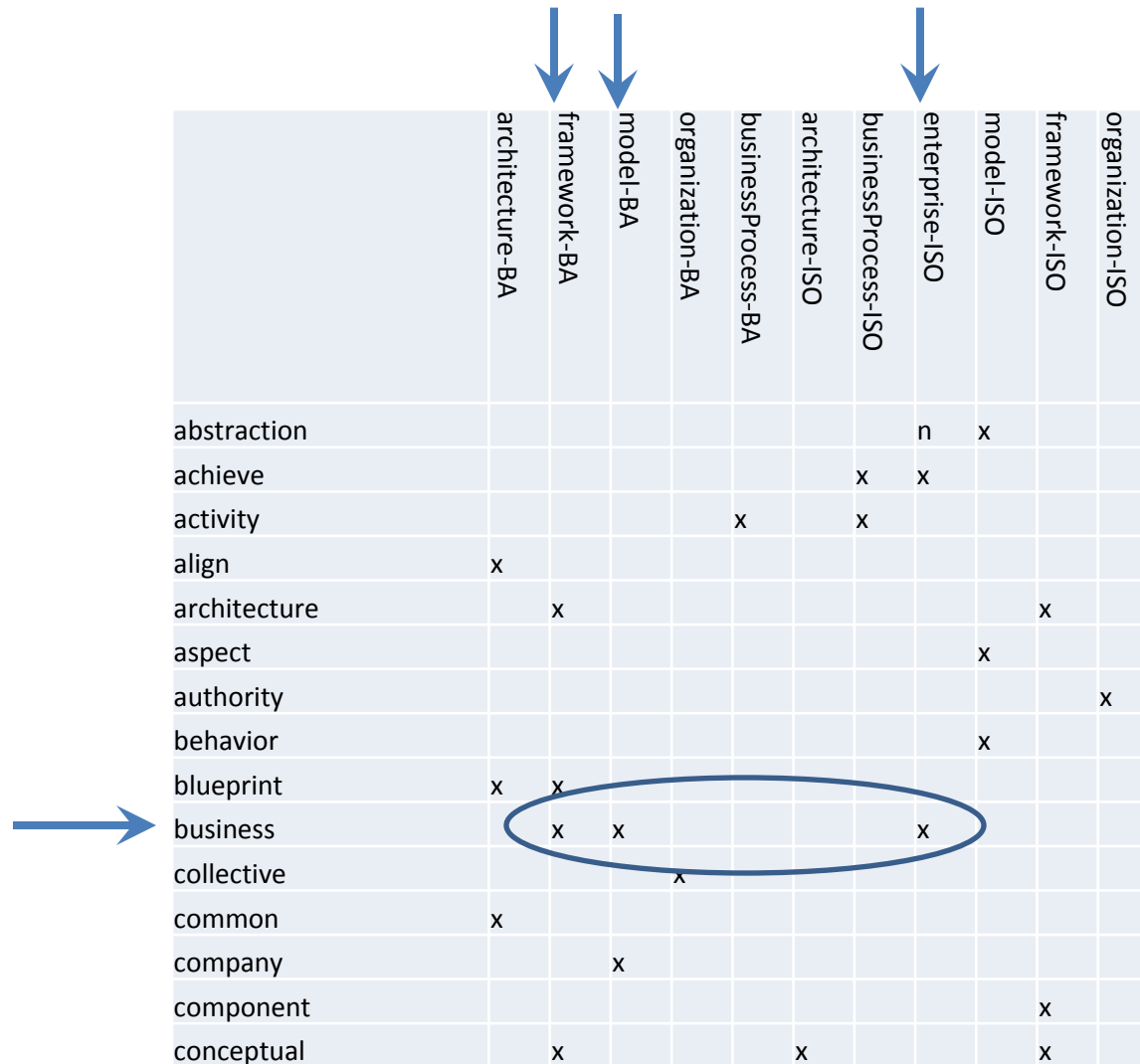
term := architecture

descriptors := business, blueprint, enterprise, provide, common, understand, organization, use, align, strategic, objective, tactical, demand

(note that 'business' is considered a descriptor)

Example FCA Context for ISO and Guild/BAS (BA)

(part of 89 attributes)



	architecture-BA	framework-BA	model-BA	organization-BA	businessProcess-BA	architecture-ISO	businessProcess-ISO	enterprise-ISO	model-ISO	framework-ISO	organization-ISO
abstraction								n	x		
achieve							x	x			
activity					x		x				
align	x										
architecture		x								x	
aspect									x		
authority											x
behavior									x		
blueprint	x	x									
business	x	x	x					x			
collective				x							
common	x										
company			x								
component										x	
conceptual		x				x				x	

FEAPO Submitted FCA Terms

- INCOSE: context, constraint, enterprise, environment, life-cycle, organization, requirement, stakeholder, system, system-of-interest, engineering, plan
- IEEE: architecture, architecture framework, architecture view, architecture viewpoint, business objective, enterprise, enterprise viewpoint, organization, organizational maturity, process
- DAMA: business architecture, business process architecture, enterprise architecture; process architecture, classification framework, enterprise data architecture, enterprise model, information system planning, Zachman framework for enterprise architecture, extended intelligent enterprise architecture
- Guild/BAS: business architecture, business blueprint, business model, business unit, capability, heat map, information concept, organization, product, strategy, stakeholder, value stream

We need your FMO's terms

- Initial analysis showed positive results
 - Identified common conceptual basis for EA
 - Identified organization-based EA concept differences
- Please get your FMO perspective included in data-set for analysis!
- For any questions, comments, and delivery of spreadsheet, contact the Lexicon deliverable owner – Richard Martin
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